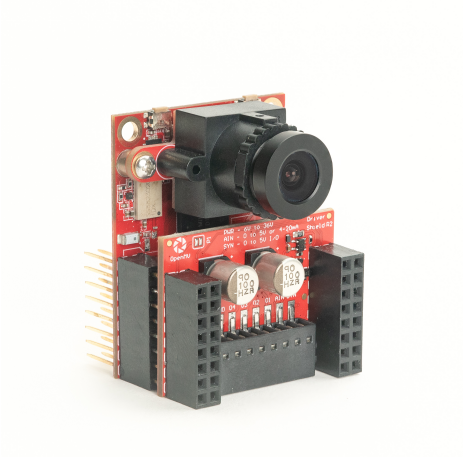
OpenMV® Driver Shield

Datasheet

SKU: OMV-SHIELD-DRIVER

Description

The Driver Shield gives your OpenMV Cam the ability to drive two 3 A motors or four 1.5 A independent channels at 6\u201336 V. It can drive 12 V or 24 V motors, solenoids, or relays. P7/P8 controls PWM for the two motor channels, while P0 and P2 control direction (or P1 via solder bridge). In independent mode, P0/P2/P7/P8/(P1) each control a high-power digital output. Includes 6\u201336 V wide-range input, robust ADC, and robust trigger I/O.



Target Areas

Motor control, robotics, solenoid actuation, relay driving, conveyor systems, industrial automation, actuator control.

Key Features

- **Dual motor driver** — Two 3 A motor channels or four 1.5 A independent channels at 6–36 V.
- **12/24 V capable** — Drive 12 V or 24 V motors, solenoids, or relays directly.
- **Independent mode** — Solder bridge configurable for 4 independent high-power digital outputs.
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.
- **Robust ADC input** — 0–5 V on P6 with ±36 V overvoltage protection. Optional 4–20 mA mode.
- **Robust trigger I/O** — 0–5 V on P10 via SYN pin.

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1 Features

1.1 Motor Driver

Component	Details
Motor Mode	Two motor channels at up to 3 A each. P7/P8 for PWM speed control, P0/P2 for direction (P1 optional via solder bridge). Operating voltage 6–36 V.
Independent Mode	Four independent channels at up to 1.5 A each. P0/P2/P7/P8/(P1) each act as high-power digital outputs. Configurable via solder bridge.
Disconnectable	All pins can be disconnected from the OpenMV Cam.

1.2 Power Input

Component	Details
Wide-Range Input	6–36 V with reverse voltage and transient surge protection. Powers motors and outputs 5.4 V on VIN via OR'ing diode.

1.3 ADC Input

Component	Details
AIN	Robust 0–5 V ADC input on P6. Overvoltage protection to ±36 V. Optional 4–20 mA mode. Disconnectable.

1.4 Digital I/O

Component	Details
SYN / P10	Robust 0–5 V digital I/O. Tolerant to -12 V, shorts to GND and 12 V. Configurable input/output. Disconnectable.

1.5 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	10 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/driver-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.